

Assignment 4 - Normalization

Yi Wang | CSCI-UA.0060 | Exercise 1: Parts Order Form | Original data and 1NF

Notation: **primary-key fields**; *foreign-key fields in 3NF*; **fields serving both roles**. Every component of a composite key is marked.

Assumptions and interpretation

- A customer number identifies one customer, with one current name and customer type. A part number identifies one part, its name, type and current unit price. Prices are treated as fixed in this data model; historical price changes are outside its scope.
- One customer places at most one order at a given date and time. Thus (customerNumber, orderDate, orderTime) identifies an order. Date and time alone need not be unique across customers.
- An order is handled by one employee. Employees can serve any customers; neither customerNumber -> employee nor employee -> customerNumber is assumed.
- A part appears at most once per order, with quantityOrdered recording its total quantity on that order. Therefore the order key plus partNumber identifies an order line.
- Assume each part type is assigned to one cage: partType -> cageCode. The example is consistent with this, but does not prove it. A cage need not identify a unique type.
- Names are stored as single values. Date/Time are renamed orderDate/orderTime, and the line-item Name/Type become partName/partType. These clarify existing fields; no attributes or surrogate keys are added.

order_form - all source data in one table (1NF)

customer Name	<u>customer Number</u>	customer Type	<u>order Date</u>	<u>order Time</u>	employee	<u>part Number</u>	part Name	part Type	cage Code	quantity Ordered	unit Price
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30am	D. Harrison	10456	Modulator	Electrical	H433	3	7
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

Each row represents one ordered part. Flattening the form's repeating parts section repeats the header values on each line and leaves one value in each cell. Under the stated uniqueness assumptions, the marked composite key identifies every row, so this table is in 1NF. The company heading is a fixed document label, not a transaction attribute.

1. Parts Order Form - 2NF

Functional dependencies and removal of partial dependencies

Dependencies used

- customerNumber -> customerName, customerType
- (customerNumber, orderDate, orderTime) -> employee
- partNumber -> partName, partType, unitPrice; partType -> cageCode
- (customerNumber, orderDate, orderTime, partNumber) -> quantityOrdered

The original key has four components. Customer details depend on customerNumber alone; employee depends on the order-level subset; part details depend on partNumber alone. These are partial dependencies, so the original table is not in 2NF. Move each set of facts to the relation determined by its key. Cage code moves with the part facts at this stage.

customer

<u>customer Number</u>	customer Name	customer Type
HG54587	Jeff Peterson	Consumer

sales_order

<u>customer Number</u>	<u>order Date</u>	<u>order Time</u>	employee
HG54587	7/1/2024	10:30am	D. Harrison

part

<u>part Number</u>	part Name	part Type	cage Code	unit Price
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order_line

<u>customer Number</u>	<u>order Date</u>	<u>order Time</u>	<u>part Number</u>	quantity Ordered
HG54587	7/1/2024	10:30am	10654	4
HG54587	7/1/2024	10:30am	10456	3
HG54587	7/1/2024	10:30am	10776	7
HG54587	7/1/2024	10:30am	10657	5

2NF check: quantityOrdered depends on the whole order-line key, and employee depends on the whole sales-order key. customer and part each have a single-attribute key. However, part is not yet in 3NF: partNumber -> partType -> cageCode.

1. Parts Order Form - 3NF

Remove the transitive dependency; retain all source information

Move cageCode from part into part_type, keyed by partType. Keep partType in part as the foreign key. The other 2NF relations already satisfy 3NF and retain their attributes.

customer

<u>customer Number</u>	customer Name	customer Type
HG54587	Jeff Peterson	Consumer

sales_order

<u>customer Number</u>	<u>order Date</u>	<u>order Time</u>	employee
HG54587	7/1/2024	10:30am	D. Harrison

part_type

<u>part Type</u>	cage Code
Plumbing	G413
Electrical	H433

part

<u>part Number</u>	part Name	part Type	unit Price
10654	Float Control	Plumbing	12
10456	Modulator	Electrical	7
10776	Hose Assembly	Plumbing	9
10657	Float Assembly	Plumbing	10

order_line

<u>customer Number</u>	<u>order Date</u>	<u>order Time</u>	<u>part Number</u>	quantity Ordered
HG54587	7/1/2024	10:30am	10654	4
HG54587	7/1/2024	10:30am	10456	3
HG54587	7/1/2024	10:30am	10776	7
HG54587	7/1/2024	10:30am	10657	5

Foreign-key references

- sales_order.customerNumber -> customer.customerNumber
- order_line.(customerNumber, orderDate, orderTime) -> sales_order.(customerNumber, orderDate, orderTime) **as one composite FK**
- order_line.partNumber -> part.partNumber; part.partType -> part_type.partType

3NF check: customer facts depend on customerNumber; employee on the full order key; quantity on the full line key; partName, partType and unitPrice on partNumber; and cageCode on partType. No non-key attribute has a remaining partial or transitive dependency under the stated assumptions. No new fields were introduced.

Assumption boundary: if a part type may occupy multiple cages, partType -> cageCode is invalid. With one cage per part, retain cageCode in part and omit part_type; that alternative is already 3NF under partNumber -> cageCode. The displayed design uses the explicit one-cage-per-type assumption.

2. Therapist Appointments - 1NF

Panacea Mental Health Corporation | Original data, assumptions and dependencies

Business rules and assumptions

- Given: a therapist may work at different branches, but on a given day sees patients at only one branch. A patient may have several appointments in one day and with different therapists.
- Assume staffNo uniquely identifies a therapist and patNo uniquely identifies a patient; each has one current name.
- Assume a therapist has at most one appointment at a particular date and start time. Appointments here are individual sessions with one patient. Hence (staffNo, appointmentDate, appointmentTime) is the chosen primary key.
- The original appointment date and time are separate atomic fields, renamed appointmentDate and appointmentTime. A recorded appointment is at that therapist's assigned branch for that day.

therapist_appointment - all source data in one table

<u>staffNo</u>	therapist Name	patNo	patName	<u>appointment Date</u>	<u>appointment Time</u>	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Initial normal form and dependencies

The table is in **1NF**: each cell has one value, and the marked key uniquely identifies a session under the assumptions. staffNo alone is insufficient because therapists have many appointments; (patNo, appointmentDate) is insufficient because patients may have several appointments that day.

- staffNo -> therapistName; patNo -> patName
- (staffNo, appointmentDate) -> branchNo
- (staffNo, appointmentDate, appointmentTime) -> patNo

therapistName depends on staffNo, and branchNo depends on (staffNo, appointmentDate). Both determinants are proper subsets of the appointment key, so these partial dependencies violate 2NF.

2. Therapist Appointments - 2NF

Separate therapist identity and daily branch assignment from individual appointments

Move therapistName to therapist, keyed by staffNo. Move branchNo to therapist_day, keyed by (staffNo, appointmentDate). Keep patNo and patName with the appointment for now: neither is determined by a proper subset of the chosen appointment key under the stated rules.

therapist

<u>staffNo</u>	therapist Name
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapist_day

<u>staffNo</u>	<u>appointment Date</u>	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

One row records a therapist's branch on a particular date. Different therapists may work at the same branch, and one therapist may use different branches on different dates.

Why this is 2NF but not yet fully 3NF

therapist has a single-field key. In therapist_day, branchNo depends on both the therapist and date. In appointment, patNo and patName have no dependency on only part of its key. All three relations therefore satisfy 2NF.

appointment still has the transitive dependency:

(staffNo, appointmentDate, appointmentTime) -> patNo -> patName.

The patient's name describes the patient, not the appointment itself. Keeping it here repeats patient names across appointments and creates an update anomaly.

appointment

<u>staffNo</u>	<u>appointment Date</u>	<u>appointment Time</u>	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

The key retains all three fields. Multiple times per day and appointments with different therapists remain possible.

2. Therapist Appointments - 3NF

Separate patient identity; preserve the therapist-date branch rule

Move patName into patient, keyed by patNo. Retain patNo in appointment as a foreign key. therapist and therapist_day already satisfy 3NF and need no further decomposition.

therapist

<u>staffNo</u>	therapist Name
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

patient

<u>patNo</u>	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

therapist_day

<u>staffNo</u>	<u>appointment Date</u>	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment

<u>staffNo</u>	<u>appointment Date</u>	<u>appointment Time</u>	patNo
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

Foreign-key references

- therapist_day.staffNo -> therapist.staffNo
- appointment.(staffNo, appointmentDate) -> therapist_day.(staffNo, appointmentDate) **as one composite FK**
- appointment.patNo -> patient.patNo

3NF check: therapistName depends only on staffNo; patName only on patNo; branchNo on the full therapist-day key; and patNo on the full appointment key. A join through therapist_day retrieves the branch for each appointment without repeating branchNo in every appointment.

branchNo is retained as a branch identifier. No branch relation is introduced because the source supplies no additional branch attributes; branchNo is therefore not marked as an FK in this design. A therapist's name and a patient's name are not assumed unique.

3. Event Staff - 1NF

Maid Better temp agency | Original data, assumptions and dependencies

Business rules and assumptions

- Given: eNo is unique to each employee. Each contract applies to one event, while an event may have multiple contracts for different service needs.
- Assume contractNo uniquely identifies a contract and eventNo uniquely identifies an event. Each employee has one recorded name; each event has one recorded location.
- Assume one record per employee-contract pair. hours is the employee's total hours for that contract, not a separate daily time entry. Employees may work on several contracts and a contract may use several employees.
- No uniqueness is assumed for names or locations. Two events can share a location, and hours need not be the same for different employees on a contract.

event_staff - all source data in one table

<u>eNo</u>	<u>contractNo</u>	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Initial normal form and dependencies

The table is in **1NF**: values are atomic and (eNo, contractNo) identifies each row. Neither component alone is a key: eNo 1135 occurs on several contracts, and C1024 includes more than one employee.

- eNo -> eName
- contractNo -> eventNo; eventNo -> eventLoc
- (eNo, contractNo) -> hours

eName depends on eNo alone. eventNo and, indirectly, eventLoc depend on contractNo alone. These are dependencies on proper subsets of the composite key, so the table is not in 2NF.

3. Event Staff - 2NF

Separate employee facts and contract facts from hours worked

Move eName into employee, keyed by eNo. Move eventNo and eventLoc into contract, keyed by contractNo. Keep hours in work_assignment because its value depends on the employee-contract combination.

employee

<u>eNo</u>	eName
1135	Smith J
1057	Hocine D
1068	White T

contract

<u>contractNo</u>	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

work_assignment

<u>eNo</u>	<u>contractNo</u>	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

Why this is 2NF but not yet fully 3NF

employee and contract each have a single-field primary key, so there are no partial dependencies on those keys. In work_assignment, hours depends on both eNo and contractNo. Thus all three relations satisfy 2NF.

contract still contains the transitive dependency:

contractNo -> eventNo -> eventLoc.

For example, H25 and its location Queens appear under both C1024 and C1026. If the location needed correction, multiple contract rows would need updating. The location belongs with the event.

3. Event Staff - 3NF

Separate event location from contracts; preserve employee hours per contract

Move eventLoc into event, keyed by eventNo. Keep eventNo in contract as a foreign key. employee and work_assignment already satisfy 3NF and retain their attributes.

employee

<u>eNo</u>	eName
1135	Smith J
1057	Hocine D
1068	White T

event

<u>eventNo</u>	eventLoc
H25	Queens
H4	Yonkers

contract

<u>contractNo</u>	eventNo
C1024	H25
C1025	H4
C1026	H25

work_assignment

<u>eNo</u>	<u>contractNo</u>	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

Foreign-key references

- work_assignment.eNo -> employee.eNo
- work_assignment.contractNo -> contract.contractNo
- contract.eventNo -> event.eventNo

3NF check: eName depends on eNo, eventNo on contractNo, eventLoc on eventNo, and hours on the full (eNo, contractNo) key. No partial or transitive dependency remains within any relation under the stated assumptions.

The final design permits multiple contracts for the same event: C1024 and C1026 both reference H25. It also preserves the many-to-many employee-contract relationship through work_assignment. Joining these four relations reconstructs all five original records, including their hours.